Lecture 6

- Conditional Expectation
- Jointly Gaussian RVs
- Gaussian Estimation
- Linear Gaussian SS model
- Kalman filter

Recall expectation and moments

theoretical ($p(x)$ known)	empirical ($X_i \sim p(x)$ i.i.d., $p(x)$ not known)
$E[X] = \int_x x p(x) dx$	$\longleftrightarrow \quad \frac{1}{N}\sum_{i=1}^{N} X_i$
$E[f(X)] = \int_x f(x) p(x) dx$	$\longleftrightarrow \quad \frac{1}{N}\sum_{i=1}^{N} f(X_i)$

these are linked by

(Strong) **Law of Large Numbers**

$$\frac{1}{N}\sum_{i=1}^{N} f(X_i) \xrightarrow{a.s.} E[f(X)] \quad \text{as } N \to \infty$$

almost sure convergence

$$Pr\left(\lim_{N\to\infty} LHS = RHS\right) = 1$$

Condition Expectation

$$E[X \mid Y = y] := \int_x x\, p(x|y)\, dx$$

"what do I expect X to be given I know $Y = y$?"

in general

$$E[f(X) \mid Y = y] := \int_x f(x)\, p(x|y)\, dx$$

we write $\mu_{x|y} := E[X \mid Y = y]$

$$\Sigma_{x|y} := Cov(X \mid Y = y) = E[(X - \mu_{x|y})(X - \mu_{x|y})^T \mid Y = y]$$

Cross Covariance

Consider $(X, Y) \sim p(x, y)$

let $Z = \begin{bmatrix} X \\ Y \end{bmatrix} \sim p(z)$

$$\mu_z = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \quad \Sigma_z = \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix} \quad \text{(blocks)}$$

where $\Sigma_{xy} := E[(X - \mu_x)(Y - \mu_y)^T]$ called cross-covariance

notice if $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$

then $\Sigma_{xy} \in \mathbb{R}^{n \times m}$

$\Sigma_{xy}^T = \Sigma_{yx} \in \mathbb{R}^{m \times n}$

proof: $\Sigma_z = E[(Z - \mu_z)(Z - \mu_z)^T]$

$$= E\left[\begin{bmatrix} X - \mu_x \\ Y - \mu_y \end{bmatrix}\begin{bmatrix} X - \mu_x \\ Y - \mu_y \end{bmatrix}^T\right]$$

$$= E\begin{bmatrix} \underbrace{(X - \mu_x)(X - \mu_x)^T}_{\Sigma_x} & \underbrace{(X - \mu_x)(Y - \mu_y)^T}_{\Sigma_{xy}} \\ \underbrace{(Y - \mu_y)(X - \mu_x)^T}_{\Sigma_{yx}} & \underbrace{(Y - \mu_y)(Y - \mu_y)^T}_{\Sigma_y} \end{bmatrix}$$

□

Jointly Gaussian R.V.s

$(X, Y) = Z \sim N(\mu_z, \Sigma_z)$

then we call (X, Y) jointly Gaussian.

Claim: as we saw

$$\mu_z = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix} \quad \Sigma_z = \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix}$$

$X \sim N(\mu_x, \Sigma_x)$ $\quad Y \sim N(\mu_y, \Sigma_y)$

Q: Given $Y = y$ and knowing (μ_z, Σ_z), how to compute posterior: $p(x|y)$?

claim: posterior is Gaussian with $p(x|y) = N(\mu_{x|y}, \Sigma_{x|y})$ where $\mu_{x|y}$ and $\Sigma_{x|y}$ come from

Gaussian Estimation Eqns

$$\boxed{\begin{aligned} \mu_{x|y} &= \mu_x + \Sigma_{xy}\Sigma_y^{-1}(y - \mu_y) \\ \Sigma_{x|y} &= \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx} \end{aligned}}$$

proof:

known: $p(x,y) = p(x|y)p(y)$ — claim:

we know LHS:

$$\eta\, e^{-\frac{1}{2}\begin{bmatrix} x-\mu_x \\ y-\mu_y \end{bmatrix}^T \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix}^{-1} \begin{bmatrix} x-\mu_x \\ y-\mu_y \end{bmatrix}}$$

want to show $= \eta_{x|y}\, e^{-\frac{1}{2}(x-\mu_{x|y})^T \Sigma_{x|y}^{-1}(x-\mu_{x|y})}$ $p(x|y)$ from Gauss Est Eqn.

$\cdot\, \eta_y\, e^{-\frac{1}{2}(y-\mu_y)^T \Sigma_y^{-1}(y-\mu_y)}$ $p(y)$

$$= \eta_{x|y}\eta_y\, e^{-\frac{1}{2}\begin{bmatrix} x-\mu_{x|y} \\ y-\mu_y \end{bmatrix}^T \begin{bmatrix} \Sigma_{x|y} & 0 \\ 0 & \Sigma_y \end{bmatrix}^{-1} \begin{bmatrix} x-\mu_{x|y} \\ y-\mu_y \end{bmatrix}}$$

now we need to show

(i) $\eta = \eta_{x|y}\eta_y$ normalization

(ii) $\begin{bmatrix} x-\mu_x \\ y-\mu_y \end{bmatrix}^T \underbrace{\begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix}^{-1}}_{\Sigma^{-1}} \begin{bmatrix} x-\mu_x \\ y-\mu_y \end{bmatrix}$

$$= \begin{bmatrix} x-\mu_{x|y} \\ y-\mu_y \end{bmatrix}^T \underbrace{\begin{bmatrix} \Sigma_{x|y} & 0 \\ 0 & \Sigma_y \end{bmatrix}^{-1}}_{D} \begin{bmatrix} x-\mu_{x|y} \\ y-\mu_y \end{bmatrix}$$

exponents

we'll prove (ii)

leave (i) as exercise for you.

Let's block diagonalize Σ as follows

$$U\Sigma L = D$$

(U: upper block diag; L: lower block diag; D: block diag)

then we have:

$$(U\Sigma L)^{-1} = D^{-1}$$

$$L^{-1}\Sigma^{-1}U^{-1} = D^{-1}$$

$$\Sigma^{-1} = LD^{-1}U$$

so

$$\begin{bmatrix} x-\mu_x \\ y-\mu_y \end{bmatrix}^T \Sigma^{-1} \begin{bmatrix} x-\mu_x \\ y-\mu_y \end{bmatrix} = \left(\begin{bmatrix} x-\mu_x \\ y-\mu_y \end{bmatrix}^T L\right) D^{-1} \left(U \begin{bmatrix} x-\mu_x \\ y-\mu_y \end{bmatrix}\right)$$

$$= \begin{bmatrix} x-\mu_{x|y} \\ y-\mu_y \end{bmatrix}^T \qquad = \begin{bmatrix} x-\mu_{x|y} \\ y-\mu_y \end{bmatrix}$$

Let's find L and U. First L:

Consider ΣL

$$\underbrace{\begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} I & 0 \\ -\Sigma_y^{-1}\Sigma_{yx} & I \end{bmatrix}}_{L} = \begin{bmatrix} \overbrace{(\Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx})}^{=\Sigma_{x|y}} & \Sigma_{xy} \\ (\Sigma_{yx} - \Sigma_y\Sigma_y^{-1}\Sigma_{yx}) & \Sigma_y \end{bmatrix}$$

L comes from Woodbury Matrix Inverse Identity

$$\Sigma L = \begin{bmatrix} \Sigma_{x|y} & \Sigma_{xy} \\ 0 & \Sigma_y \end{bmatrix}$$

now for U:

$$U(\Sigma L) = \begin{bmatrix} I & -\Sigma_{xy}\Sigma_y^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} \Sigma_{x|y} & \Sigma_{xy} \\ 0 & \Sigma_y \end{bmatrix}$$

$$= \begin{bmatrix} \Sigma_{x|y} & (\Sigma_{xy} - \Sigma_{xy}\Sigma_y^{-1}\Sigma_y) \\ 0 & \Sigma_y \end{bmatrix}$$

we know:

$$U\Sigma L = \begin{bmatrix} \Sigma_{x|y} & 0 \\ 0 & \Sigma_y \end{bmatrix}$$

now find:

$$\left(U \begin{bmatrix} x-\mu_x \\ y-\mu_y \end{bmatrix}\right)$$

$$= \begin{bmatrix} I & -\Sigma_{xy}\Sigma_y^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} (x-\mu_x) \\ (y-\mu_y) \end{bmatrix}$$

$$= \begin{bmatrix} (x-\mu_x) - \Sigma_{xy}\Sigma_y^{-1}(y-\mu_y) \\ (y-\mu_y) \end{bmatrix}$$

$$= \begin{bmatrix} x - \overbrace{(\mu_x + \Sigma_{xy}\Sigma_y^{-1}(y-\mu_y))}^{\mu_{x|y}} \\ y-\mu_y \end{bmatrix}$$

we proved:

$$\boxed{\begin{aligned} \Sigma_{x|y} &= \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx} \\ \mu_{x|y} &= \mu_x + \Sigma_{xy}\Sigma_y^{-1}(y-\mu_y) \end{aligned}}$$

□

Q: But how to relate this to more familiar dynamical / algebraic system models?

Recall Linear SS model:

$$x_{t+1} = Ax_t + Bu_t$$

$$y_t = Cx_t + Du_t$$

Linear Gaussian SS Model:

dynamics model $X_{t+1} = A_t X_t + B_t u_t + W_t$ (process noise covariance)

where $W_t \sim GWN(0, Q_t)$

sensor model $Y_t = C_t X_t + D_t u_t + V_t$

where $V_t \sim GWN(0, R_t)$

means Gaussian White Noise (GWN) (more details on this later)

R_t: measurement noise covariance

Claim: This model

(i) Gives state and measurement sequence $(X_{0:t}, Y_{1:t})$ that is jointly Gaussian

(ii) Is Markov

(iii) Has conditionally independent measurements.

- Recall, (i) and (iii) are the requirements for the recursive Bayesian filter.
- The special case of the recursive Bayesian filter for the Linear Gaussian State Space model is the Kalman Filter.

We first define notation

prediction distribution

$$p(x_t | y_{1:t-1}) = N(\mu_{t|t-1}, \Sigma_{t|t-1})$$

where

$$\mu_{t|t-1} := E[X_t | Y_{0:t-1}]$$

and $$\Sigma_{t|t-1} := Cov(X_t | Y_{0:t-1})$$

update distribution

$$p(x_t | y_{1:t}) = N(\mu_{t|t}, \Sigma_{t|t})$$

where

$$\mu_{t|t} := E[X_t | Y_{0:t}]$$

$$\Sigma_{t|t} := Cov(X_t | Y_{0:t})$$

Kalman Filter gives recursive expressions to efficiently compute: prediction: $\mu_{t|t-1}$, $\Sigma_{t|t-1}$ and update: $\mu_{t|t}$, $\Sigma_{t|t}$

Kalman Filter Algo.

Given:

model parameters $A_t, B_t, C_t, D_t, Q_t, R_t$ and prior $X_0 \sim N(\mu_{0|0}, \Sigma_{0|0})$

predict:

mean: $\mu_{t|t-1} = A_{t-1}\mu_{t-1|t-1} + B_{t-1}u_{t-1}$

cov: $\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1}$

update:

mean: $\mu_{t|t} = \mu_{t|t-1} + K_t(y_t - C_t\mu_{t|t-1})$

cov: $\Sigma_{t|t} = \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1}$

where $K_t = \Sigma_{t|t-1}C_t^T(C_t\Sigma_{t|t-1}C_t^T + R_t)^{-1}$

K_t: Kalman Gain

Next time we derive this using the Gaussian Est. eqns.